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§1 微分の計算

いろいろな関数の微分

a は実数, a は $a > 0, a \neq 1$ をみたす実数とする。

$$(x^a)' = a x^{a-1}$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$\left(\frac{1}{\tan x}\right)' = -\frac{1}{\sin^2 x}$$

$$(\log_a x)' = \frac{1}{x}$$

$$(\log_a a)' = \frac{1}{x \log_a a}$$

$$(\log |x|)' = \frac{1}{x}$$

$$(\log_a |x|)' = \frac{1}{x \log_a a}$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \log a$$

k, l は定数 とする。

$$y = \cancel{k f(x)} + \cancel{l g(x)} \quad \text{ならば} \quad y' = k f'(x) + l g'(x)$$

別々に微分して良い

(例1) 次の関数を微分せよ。

(1) $y = \frac{1}{x}$

$$y' = (x^{-1})'$$

$$= -1 \cdot x^{-1-1}$$

$$= -x^{-2}$$

$$= -\frac{1}{x^2} \text{,,}$$

(2) $y = \frac{1}{2x^2}$

$$y' = \left(\frac{1}{2} x^{-2}\right)'$$

$$= \frac{1}{2} \cdot (-2) \cdot x^{-2-1}$$

$$= -x^{-3}$$

$$= -\frac{1}{x^3} \text{,,}$$

(3) $y = \sqrt{x}$

$$y' = (x^{\frac{1}{2}})'$$

$$= \frac{1}{2} x^{\frac{1}{2}-1}$$

$$= \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x}} \text{,,}$$

(4) $y = \sqrt[3]{x^2}$

$$y' = (x^{\frac{2}{3}})'$$

$$= \frac{2}{3} x^{\frac{2}{3}-1}$$

$$= \frac{2}{3} x^{-\frac{1}{3}}$$

$$= \frac{2}{3\sqrt[3]{x}} \text{,,}$$

(例2) 次の関数を微分せよ。

(1) $y = \frac{3x^3 + x^2 - 2x + 1}{x^2}$

$$y = 3x + 1 - \frac{2}{x} + \frac{1}{x^2}$$

$$= 3x + 1 - 2x^{-1} + x^{-2}$$

よ)

$$y' = 3 - 2(-1)x^{-2} - 2x^{-3}$$

$$= 3 + \frac{2}{x^2} - \frac{2}{x^3} \text{,,}$$

(2) $y = 2 \cos x - \tan x$

$$y' = 2 \cdot (-\sin x) - \frac{1}{\cos^2 x}$$

$$= -2 \sin x - \frac{1}{\cos^2 x} \text{,,}$$

(3) $y = e^x - \log x$

$$y' = e^x - \frac{1}{x} \text{,,}$$

(4) $y = 2^x - \log_2 x$

$$y' = 2^x \log 2 - \frac{1}{x \log 2} \text{,,}$$